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End Semester Examination 2024

B.Tech.: Civil Engineering

Semester: 3rd

Paper Name: Engineering Mathematics-III

Paper Code: TMA 302

Time: 3 Hours

Max. Marks: 100

Note: -

- (i) All questions are compulsory.
- (ii) Answer any two sub questions among a, b & c in each main question
- (iii) Total marks in each main question are twenty.
- (iv) Each question carries 10 marks

Q 1

20 Marks CO1

(a) Check the function $u(x, y) = x^2 - y^2 - 2xy - 4x + 7y$ is harmonic or not, if yes, then find a conjugate harmonic function $v(x, y)$.

(b) Use Cauchy's integral formula to evaluate $\int_C f(z) dz$ where $C: |z| = 3$.

Given that $f(z) = \frac{3z^2 + 7z + 1}{(z-1)(z-2)}$.

(c) Check the analyticity of a complex valued function $f(z) = \frac{1}{z}$; $z \neq 0$, where

$z = x + iy$.

Q 2

20 Marks CO1/CO2

(a) Define absolute, relative and percentage errors. Find the relative error if 0.005998 is rounded off correct to three decimal places approximated to 0.006.

(b) Define the analytic function, and hence to construct an analytic function $f(z)$ using Milne's Thomson method, where the real part of function $f(z) = u(x, y) + iv(x, y)$ is

$u(x, y) = x^3 - 3xy^3$.

(c) Using Cauchy's integral formula to evaluate the value of the integral $\oint \frac{e^{zt}}{z^2 + 1} dz$, along the closed curve $C: |z| = 2$.

Q 3**20 Marks CO3/CO4**

(a) Using iteration method to find a real root of an equation $xe^x = 1$ correct to three significant digits.

(b) Values of $f(x)$ in the interval $[0, 4]$ are given below:

x	0	1	2	3	4
f(x)	3	10	21	36	55

Using Simpson's one-third rule with a step size of 1, find the numerical approximation (rounded off to the second decimal place) of $\int_0^4 f(x)dx$.

(c) Using Newton-Raphson method to compute a real positive root of the equation $x^3 - x - 1 = 0$ with initial condition $x_0 = 1.6$, (Correct to three decimal places).

Q 4**20 Marks CO5**

(a) The first moments about the mean of a frequency distribution are 0, 7.058, 36.151, and 408.738. Calculate β_1 , β_2 , and comment upon the Skewness and Kurtosis of the distribution.

(b) If the probability density function of a continuous random variable X is given by

$$f(x) = \begin{cases} kx^4; & 0 < x < 1 \\ 0; & \text{otherwise} \end{cases}$$

Find the value of k and hence compute its mean, variance, and standard deviation.

(c) Using Least square method to fit a curve $y = ax + b$ for the following data:

x	1	3	5	10	15
y	10	5	20	25	22

Q 5**20 Marks CO6**

(a) Using Least square method to fit a curve $y = ax^2 + bx + c$ for the following data:

x	1	3	5	10	15
y	10	5	20	25	22

(b) Explain the concept of the Kurtosis, Skewness, Correlation and Regression analysis.

(c) A discrete random variable X has the probability function given below:

$X = x$	0	1	2	3	4	5	6	7	8
$P(x)$	0.01	0.03	0.05	0.07	0.09	0.11	0.13	0.15	0.17

Find the expected value and Standard deviation of random variable X .

